Ex. N0:4

Breadth First Search

AIM:

To write a program to solve any problem using Breadth First Search in SWI Prolog.

ALGORITHM:

Step 1: Start.
Step 2: Define the graph by specifying the edges between nodes and their corresponding costs.
Step 3: Initialize the source node as 0 and the target node as 9.
Step 4: Insert the source node into the priority queue with priority 0.
Step 5: Initialize the visited list with the source node.
Step 6: Select and remove the node with the lowest priority from the priority queue.
Step 7: Check whether the selected node is the target node.

- If yes, stop the search.
- If no, continue to the next step.

Step 8: Find all adjacent nodes of the selected node that have not been visited.
Step 9: Add the unvisited adjacent nodes to the visited list and insert them into the priority queue with their respective edge costs.
Step 10: Sort the priority queue in ascending order of priority.
Step 11: Repeat Steps 6–10 until the target node is found or the priority queue becomes empty.
Step 12: Display the nodes visited during the search.
Step 13: Stop.

PROGRAM:

```
edge(1, 5, 8).
edge(2, 6, 12).
edge(2, 7, 14).
edge(3, 8, 7).
edge(8, 9, 5).
edge(8, 10, 6).
edge(9, 11, 1).
edge(9, 12, 10).
edge(9, 13, 2).
adjacent(X, Y, C) :- edge(X, Y, C).
adjacent(X, Y, C) :- edge(Y, X, C).
best_first_search(Source, Target) :-
   (  search([0-Source], [Source], Target, [], VisitedOrder)
   -> write('Nodes visited: '),
      writeln(VisitedOrder)
   ;  writeln('Target not found.')
   ).
search([_-Node|_], _, Target, Acc, Path) :-
   Node =:= Target,
   !,
```

```
    reverse([Node|Acc], Path).
search([_-Node|Rest], Visited, Target, Acc, Path) :-
    findall(C-N,
        ( adjacent(Node, N, C),
          \+ memberchk(N, Visited)
        ),
        NewEntries),

    findall(N, member(_-N, NewEntries), NewNodes),

    append(Visited, NewNodes, Visited1),
    append(Rest, NewEntries, Queue0),
    sort(Queue0, Queue1),

    search(Queue1, Visited1, Target, [Node|Acc], Path).
run :-
    best_first_search(0, 9).
```

OUTPUT:



RESULT:

Thus the Breadth First Search program was executed successfully in SWI Prolog console and verified.